Name: ____________________________ Date: ________________ Period: _________

Practice

1. Consider the equation $x^2 + y^2 = 49$.

   (a) Solve for y. Write the two explicit functions $f_1(x)$ and $f_2(x)$.

   $f_1(x) =$ _______________________ $f_2(x) =$ _______________________

   (b) State the domain of each function.

   Domain: _____________________________

2. For the equation $x^2 + y^2 = 49$, complete the table for the upper half.

x	$y = f_1(x)$
-7	______________
0	______________
5	______________
7	______________

3. Two points on the upper half of $x^2 + y^2 = 49$ are $(5, \sqrt{24})$ and $(6, \sqrt{13})$. (You may use decimal approximations: $\sqrt{24} \approx 4.899$ and $\sqrt{13} \approx 3.606$.)

   (a) Compute $\Delta y/\Delta x$ between these two points.

   $\Delta y/\Delta x \approx$ _________________

   (b) Is co-variation positive or negative? What does this mean? ___________________________________

4. A budget constraint for a school supply purchase is $3x + 5y = 60$, where x is the number of notebooks and y is the number of binders purchased.

   (a) Solve for y as an explicit function of x.

   $y =$ _____________________________

   (b) As the number of notebooks x increases by 1, what happens to the number of binders y that can be purchased? By how much?

   (c) Is the co-variation between x and y positive or negative? _______________________

5. For the equation $9x^2 + y^2 = 9$, determine the co-variation sign between x and y on the upper half as x increases from 0 to 1. Use two nearby points to support your answer. (Hint: find y at $x = 0$ and $x = 0.5$.)

Co-variation sign: ________________ Reason: ________________________________

AP-Style Multiple Choice

6. The equation $x^2 - 4y^2 = 16$ is graphed in the xy-plane. Which of the following best describes the co-variation of x and y on the upper branch of the graph (where $y > 0$) as x increases from 4 to 8?

(A) Co-variation is positive because y increases as x increases.

(B) Co-variation is positive because y decreases as x increases.

(C) Co-variation is negative because y increases as x increases.

(D) Co-variation is negative because y decreases as x increases.

Extension optional

Extension (optional): Consider the equation $(x - 2)^2 + (y - 1)^2 = 25$.

1. Identify this as a circle and state its center and radius.
2. Solve for the two explicit functions.
3. Find two points on the upper half and compute $\Delta y/\Delta x$. Is co-variation positive or negative near $x = 2$?